

**Fourth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ANALOG AND DIGITAL COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.
 - (a) Derive the expression for total transmitted power in AM wave in terms of carrier power. A 400W carrier is modulated to a depth of 75%. Calculate the total power and individual side band power in the modulated wave.
5(CO1)
 - (b) An equation of frequency modulation wave given by,
 $E_{FM} = 10 \sin (10^8 t + 3 \sin 10^4 t)$. Determine carrier frequency, modulation frequency, modulation index, frequency deviation and power dissipated by FM wave for 100Ω resistance. Also write an equation of phase modulation.
5(CO1)
 2.
 - (a) Analyze the Noise effect on AM Double Side Band Suppressed Carrier system.
5(CO3)
 - (b) The signal power and noise power measured at the input of an amplifier are $150\mu W$ and $1.5\mu W$ respectively. If the signal power at the output $1.5W$ and noise power is $40mW$, then calculate noise factor and noise figure.
5(CO3)

3. (a) The pulse code modulation is used for TV signal has bandwidth of **4.2 MHz**. The number of quantization levels is 512. Determine :
 - (1) Code word length,
 - (2) Transmission bandwidth,
 - (3) Maximum bit rate and
 - (4) Output signal to quantization noise ratio in dB. 5(CO1)
- (b) Describe the delta modulation with neat block diagram. Write its advantages. 5(CO1)

4. (a) Derive an expression for Probability of errors for Amplitude Shift keying. 5(CO2)
- (b) Describe the concept of Inter symbol interference along with Nyquist criterion. 5(CO2)

5. (a) Describe the operation of Non-coherent detection of Binary FSK with neat block diagram. 5(CO4)
- (b) Draw waveforms for QPSK Transmitter if input data stream is **[1 1 0 1 0 0 0 1]**. 5(CO4)

6. (a) A Convolution Code is described by $X1 = [1 \ 1 \ 1]$; $X2 = [1 \ 0 \ 1]$; Draw the convolution encoder, code tree, code trellis and state diagram. 5(CO5)
- (b) The convolution encoded received sequence is **[1 1 0 1 1 1]**, decode the sequence using Viterbi decoding algorithm and find maximum likelihood path. 5(CO5)

